Solana SVM Study Course

Course Overview

Course Title

Mastering Solana and SVM Development: From EVM to Solana

Duration & Level

- **Duration**: 16 weeks (part-time) / 8 weeks (full-time)
- **Level**: Intermediate to Advanced
- **Prerequisites**: JavaScript/TypeScript, basic blockchain concepts, familiarity with EVM development

Learning Objectives

- Master Solana blockchain architecture and SVM (Solana Virtual Machine)
- Build Solana applications using NestJS
- Understand key differences between EVM and SVM paradigms
- Implement blockchain solutions
- Deploy and monitor Solana applications in production

Course Repository Structure

```
solana-svm-study/
├── docs/                              # Course documentation
│   ├── COURSE.md                     # This course curriculum
│   ├── STUDY.md                      # Detailed study topics
│   ├── TASKS.md                      # Implementation tasks
│   ├── PROJECT.md                    # Project overview
│   └── diagrams/                     # Architecture diagrams
├── src/                              # Application source code
│   ├── modules/                      # Feature modules
│   │   ├── accounts/                # Account management
│   │   ├── transactions/            # Transaction services
│   │   ├── tokens/                  # SPL token operations
│   │   └── ...                      # Additional modules
│   ├── common/                      # Shared utilities
│   ├── database/                    # Database layer
│   └── shared/                      # Shared types
├── infra/                            # Infrastructure as code
├── scripts/                          # Utility scripts
└── ...                               # Configuration files
```

Technology Stack

Backend Framework

- **NestJS**: Progressive Node.js framework
- **TypeScript**: Type-safe JavaScript
- **TypeORM**: TypeScript ORM for databases

Blockchain Integration

- **Solana Web3.js**: Solana blockchain interaction
- **@solana/web3.js**: Official Solana JavaScript SDK

Database & Messaging

- **PostgreSQL**: Primary database
- **Apache Kafka**: Event streaming

Development Environment

Local Development

- Docker Compose for services
- Hot reload with NestJS CLI
- Integrated testing with Jest
- Code coverage reporting

Production Deployment

- Kubernetes manifests
- CI/CD with GitHub Actions
- Multi-stage Docker builds
- Health checks and monitoring

Course Modules Overview

Core Modules

1. **Accounts & Programs** - Account management and PDAs
2. **Transactions & Instructions** - Transaction building and submission
3. **Token Standards** - SPL token operations
4. **Account Abstraction** - Smart accounts and PDAs
5. **Fee Mechanism** - Transaction fees and prioritization

Advanced Modules

6. **Consensus & Validation** - Block validation and consensus
7. **Signing & Cryptography** - Digital signatures and keys
8. **MPC** - Multi-party computation protocols
9. **SVM** - Solana Virtual Machine integration

Learning Approach

Hands-on Implementation

- Build complete NestJS API for Solana integration
- Implement all major SVM features
- Real-world blockchain application development

Progressive Complexity

- Start with basic concepts (accounts, transactions)
- Build up to advanced features (MPC, CPIs)
- End with production deployment and monitoring

EVM to SVM Migration

- Direct comparisons between EVM and SVM concepts

Assessment & Evaluation

Implementation Tasks

- Complete all module implementations
- Pass comprehensive test suites
- Code review and optimization

Project Milestones

- Database schema and migrations
- API endpoint implementation
- Integration testing
- Performance optimization

Final Deliverables

Success Metrics

Technical Proficiency

- 80%+ test coverage
- Performance benchmarks met
- Security best practices implemented
- Production deployment successful

Code Quality

- TypeScript strict mode compliance
- Clean architecture patterns
- Comprehensive error handling
- API documentation complete

Thank You!

Questions & Next Steps

- Review the STUDY.md for detailed topics
- Check TASKS.md for implementation guide
- Start with Module 1: Accounts & Programs

Happy Learning! 🚀